Setting up an OpenRouter Account, Navigating the Interface and Troubleshooting

Objective

In this practical, learners will create an OpenRouter account, explore the OpenRouter dashboard, understand where models and API keys are managed, and identify free models that can be used later with Open WebUI.

OpenRouter provides access to multiple AI models through a single platform and API key. For this course, we will use free OpenRouter models, especially models whose names end with (free). Some third-party setup guides also note that when no credits are added, learners should choose models ending in (free).

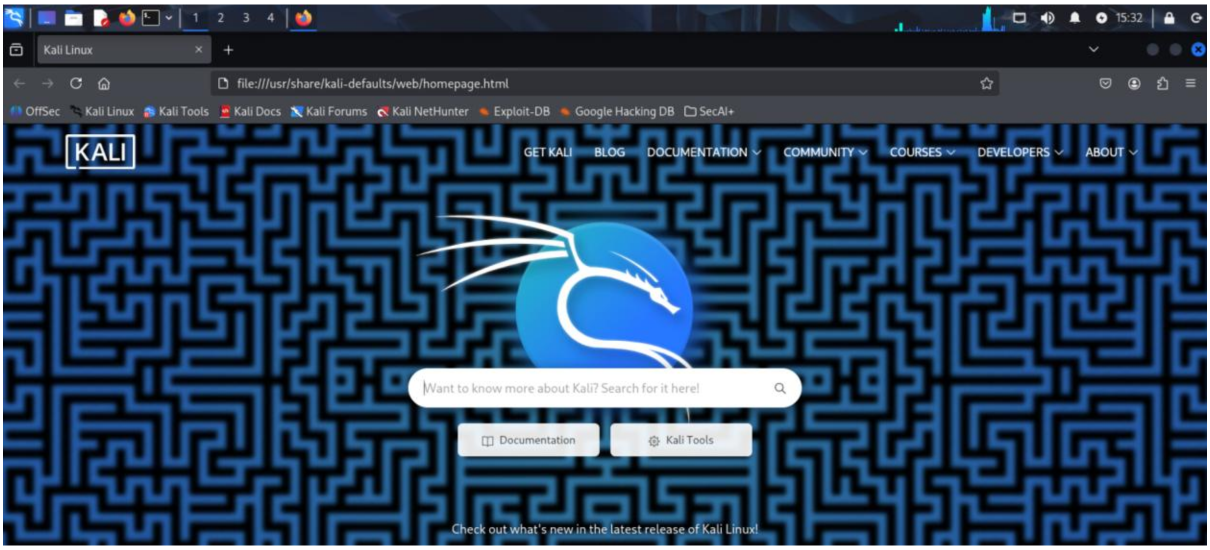
Lab Environment

Operating System: Kali Linux
Browser: Firefox
Platform: OpenRouter
Use case: AI for Cybersecurity practicals
Model type: Various models connected to OpenRouter

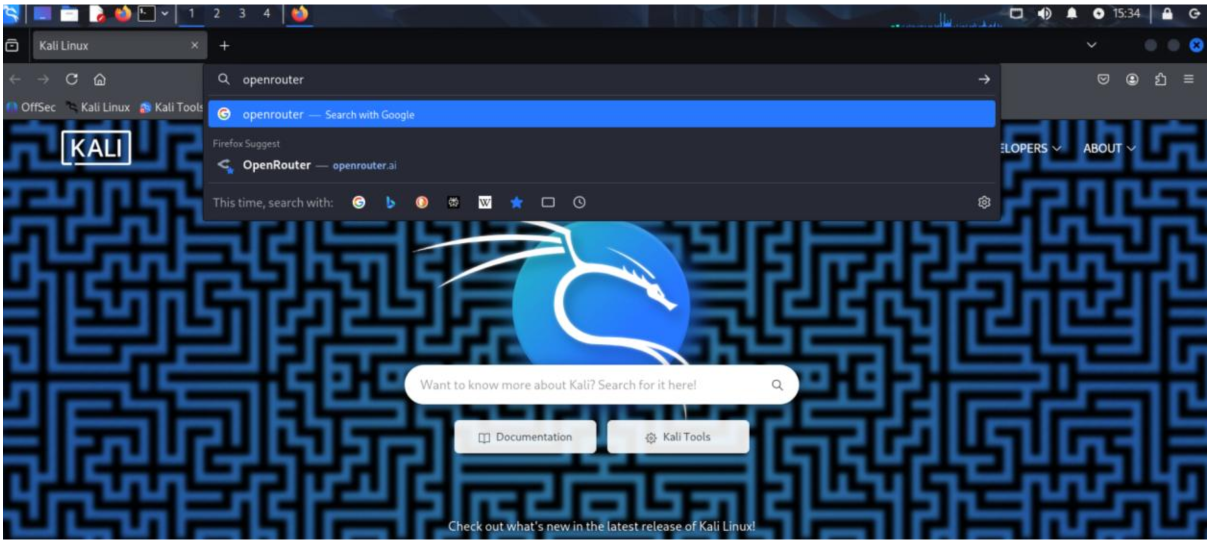
Step 1: Open Firefox in Kali Linux

1. Start your Kali Linux machine.

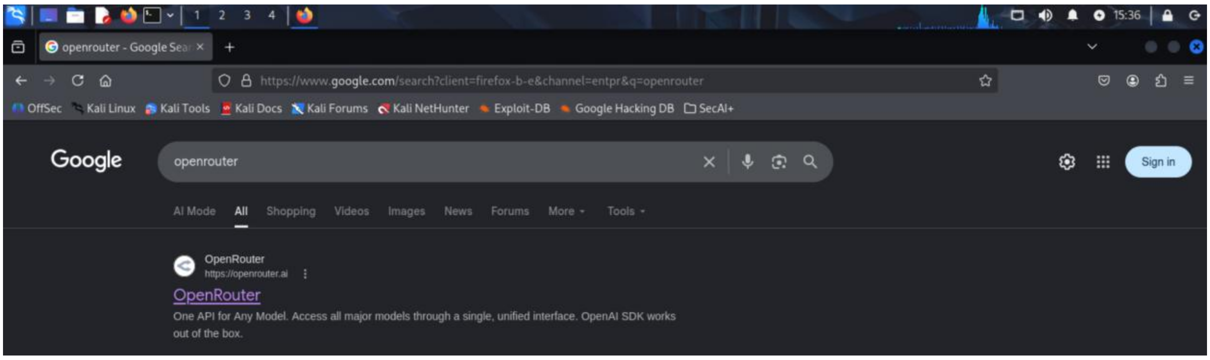
2. Open **Firefox**.



3. In the address bar, go to OpenRouter.

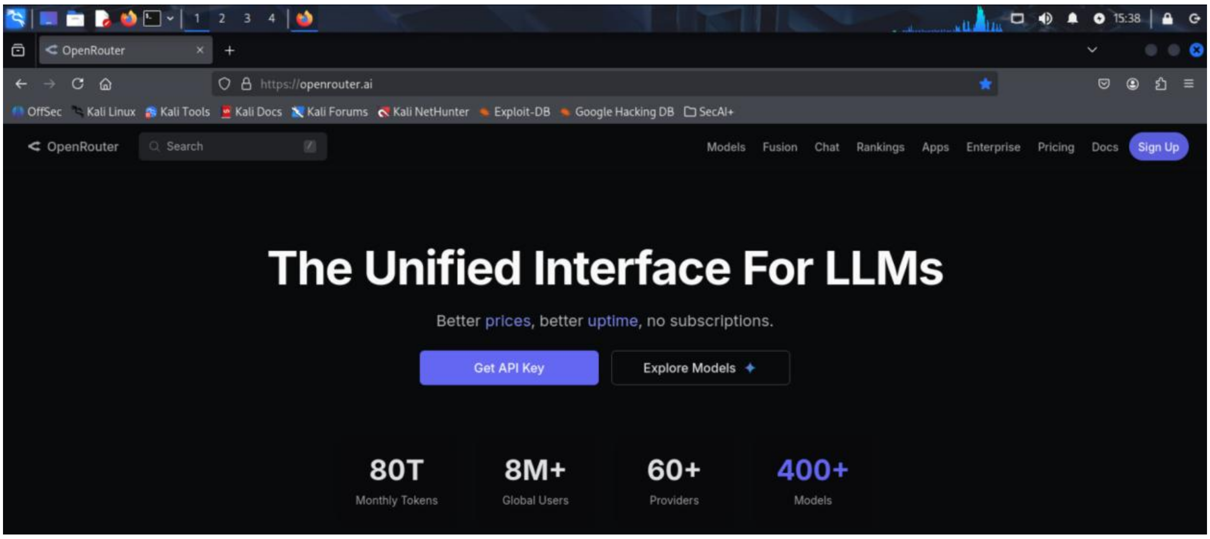


4. Select https://openrouter.ai

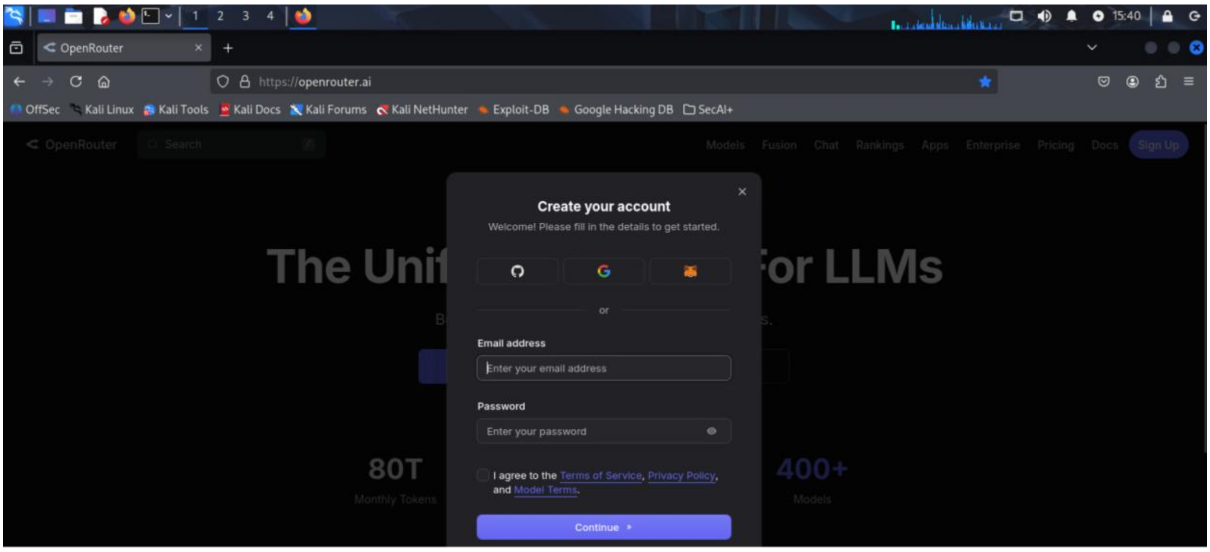


Step 2: Create or sign in to an OpenRouter Account

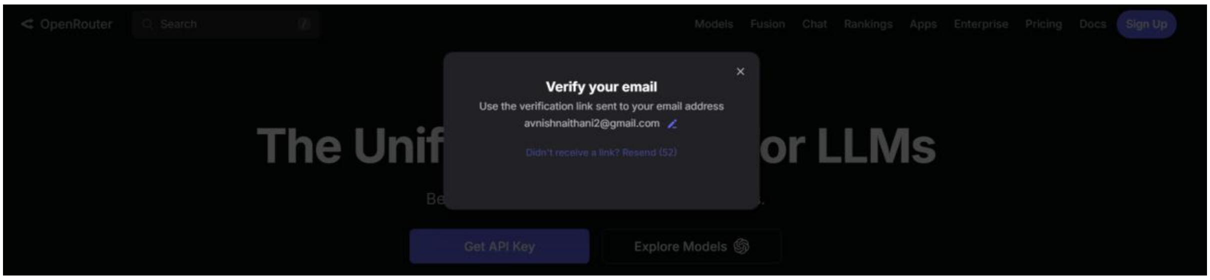
1. On the OpenRouter homepage, click **Sign up** to create an account.



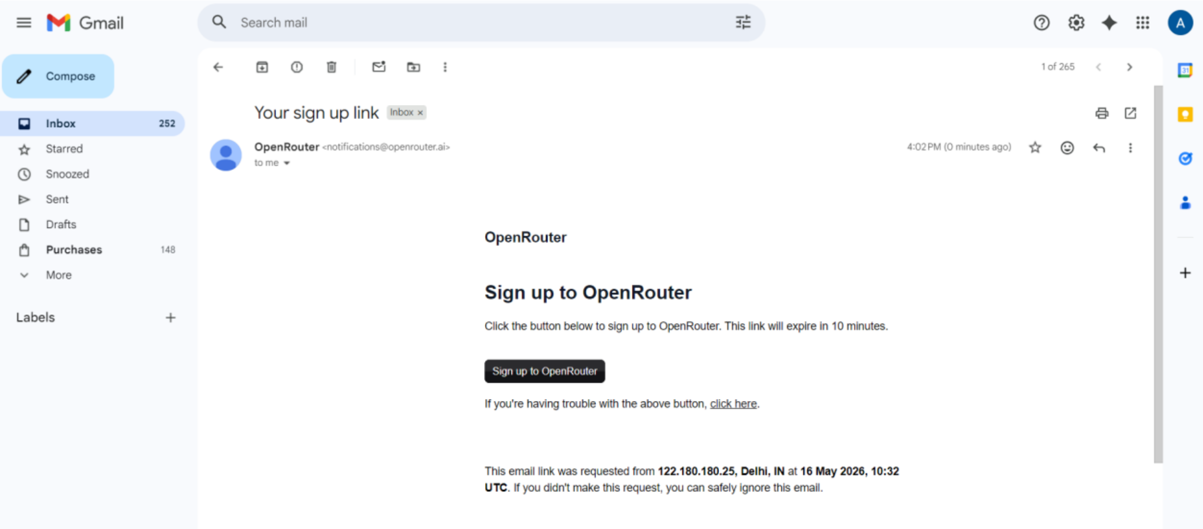
2. Type in an email address you want to Sign Up with and create a password. Click on the I agree checkbox and click on continue.



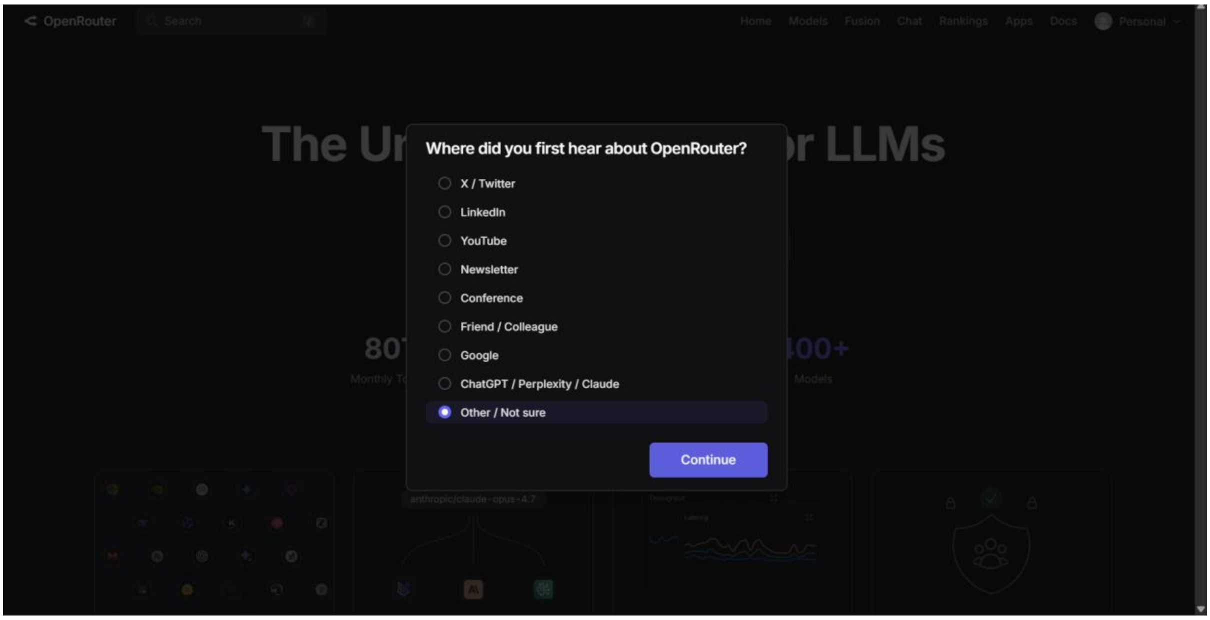
3. A verification link will be sent to your registered email.



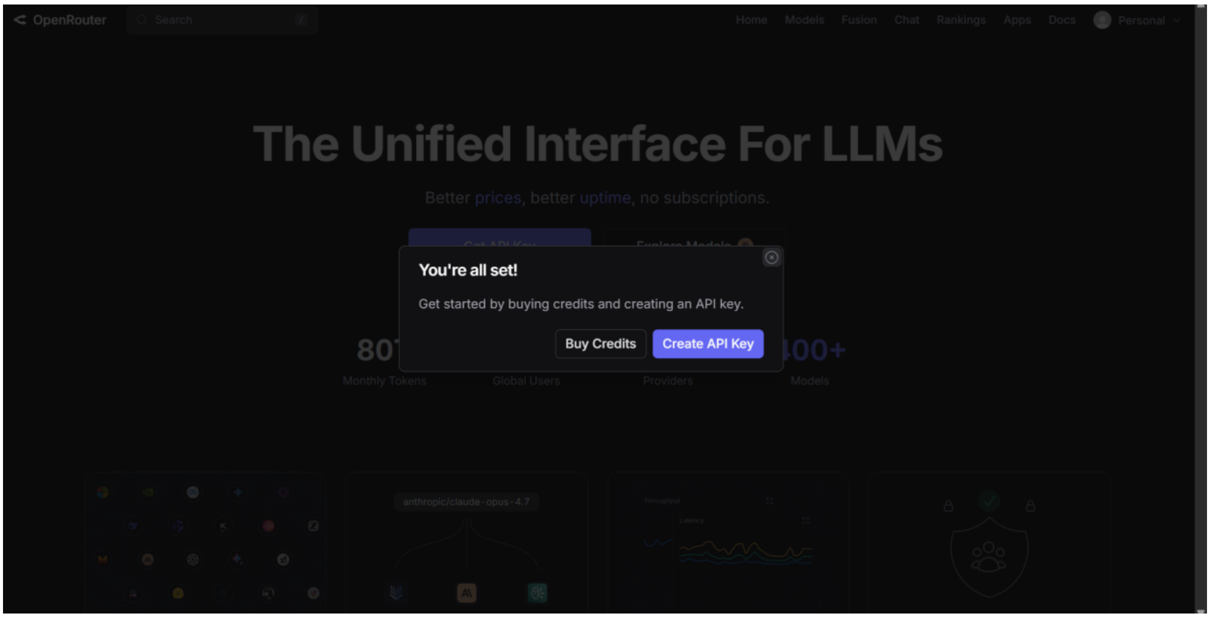
4. Go to you mailbox and click on “Sign up to OpenRouter”.



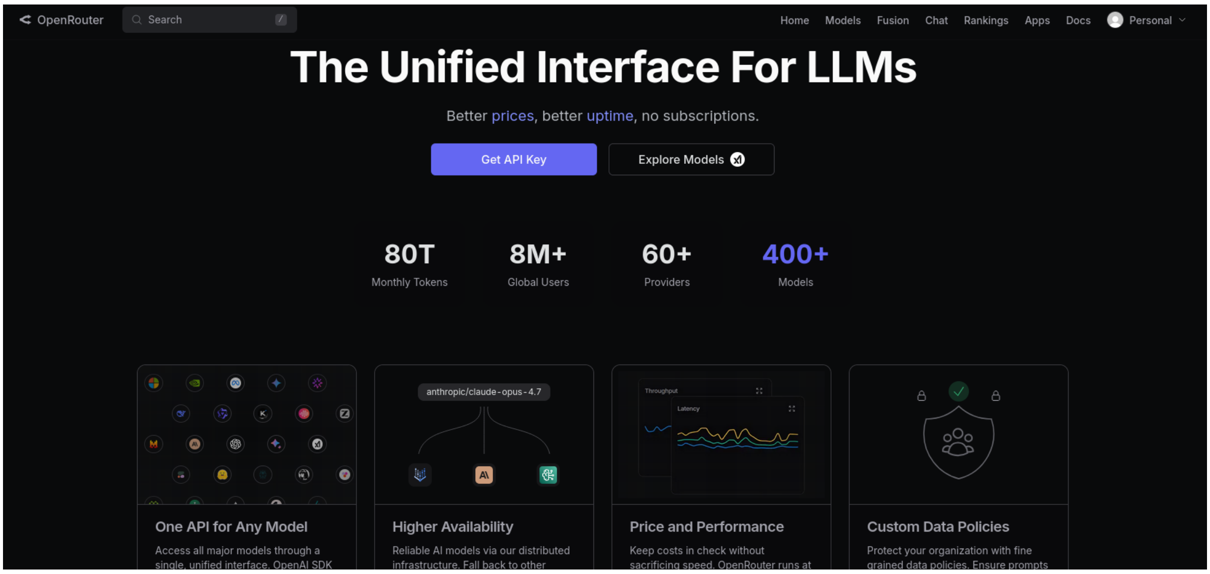
5. On doing so it will now log you into your OpenRouter account, you can close the other OpenRouter tab where you first entered you Sign Up details.



6. Click on “Continue” And close this pop up displayed.



7. You can now see the OpenRouter home page and you are all set to explore.



Step 3: Explore the OpenRouter Home Page / Dashboard

After logging in to OpenRouter, you will land on the main OpenRouter interface.

In the screenshot, the account is already signed in because the top-right corner shows the user profile as **Personal**.

What to Observe on This Page

At the top of the page, OpenRouter provides the main navigation menu.

The important navigation options visible here are:

Option	Purpose
Home	Returns to the main OpenRouter landing page
Models	Used to browse available AI models
Fusion	OpenRouter's routing/model-combination related feature
Chat	Lets users test models directly inside OpenRouter

Option	Purpose
Rankings	Shows model rankings based on usage/performance metrics
Apps	Shows apps and integrations that work with OpenRouter
Docs	Provides documentation for API usage and integration
Personal	Account/profile menu for settings, keys, credits, and usage

Action to Perform

Click **Explore Models** or the **Models** option from the top navigation bar.

This will take you to the model catalog, where we will search for free models that can be used in later AI cybersecurity practicals.

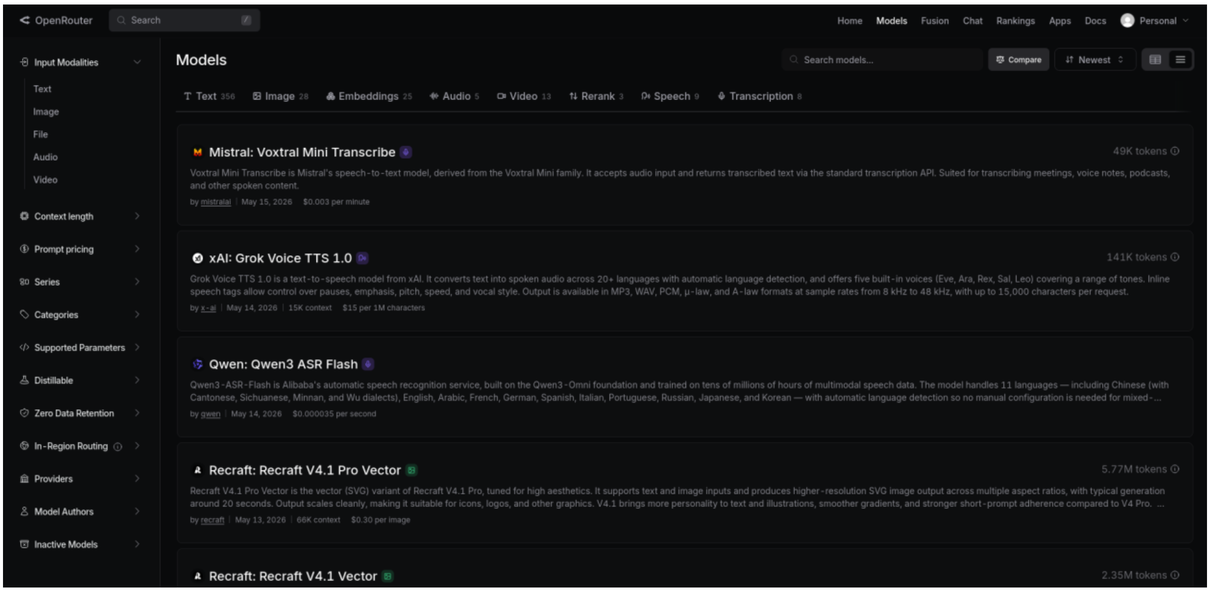
Step 4: Navigate to the Models Page

After clicking **Models** from the top navigation bar, OpenRouter opens the **Models** catalog page.

This page is used to browse, search, filter, and compare the AI models available through OpenRouter.

What to Observe

The page is divided into three main areas:



Area	Purpose
Left filter panel	Filter models by input type, context length, pricing, provider, and other properties
Model category tabs	Quickly switch between Text, Image, Embeddings, Audio, Video, Rerank, Speech, and Transcription models
Model list	Displays available models with name, description, provider, release date, pricing, and token information

The **Text** tab is visible with many available models. Other tabs such as **Image**, **Embeddings**, **Audio**, and **Video** are also available.

Important Options on the Models Page

1. Search Models

At the top-right side, there is a **Search models** box.

This is useful for finding models by name, provider, or keyword.

Example searches:

free

llama

qwen

mistral

gemma

deepseek

2. Compare

The **Compare** button allows users to compare selected models.

This can be useful later when comparing smaller and larger models for prompt-refining quality.

3. Sorting Option

The sorting dropdown currently shows **Newest**.

This means the models are being sorted by most recently added or updated.

For this lab, sorting is less important than filtering for **free text models**.

4. View Layout

The icons near the top-right corner allow changing how models are displayed.

The current view shows models in a detailed list format, which is useful because it displays model descriptions and pricing information.

Left-Side Filters

The left-side panel contains filters such as:

Filter	Use
Input Modalities	Select Text, Image, File, Audio, or Video input
Context length	Filter by how much input the model can process

Filter	Use
Prompt pricing	Filter models based on input cost
Series	Filter by model family
Categories	Filter by model type or task category
Supported Parameters	Filter based on supported API features
Providers	Filter models by hosting/provider
Model Authors	Filter by model creator
Inactive Models	View older or inactive models

Action to Perform

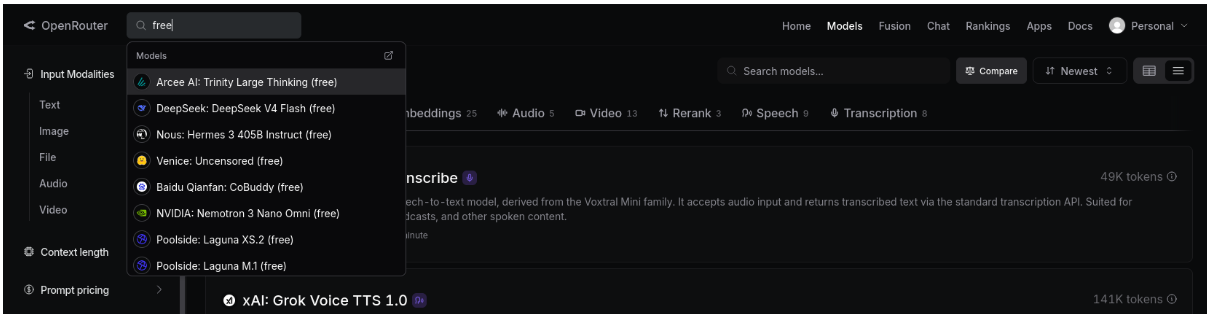
In the **Search models** box, type:

free

This will help identify models that are available for free usage.

After searching, look for models whose names or model IDs clearly indicate free usage, commonly with a suffix such as:

(free)



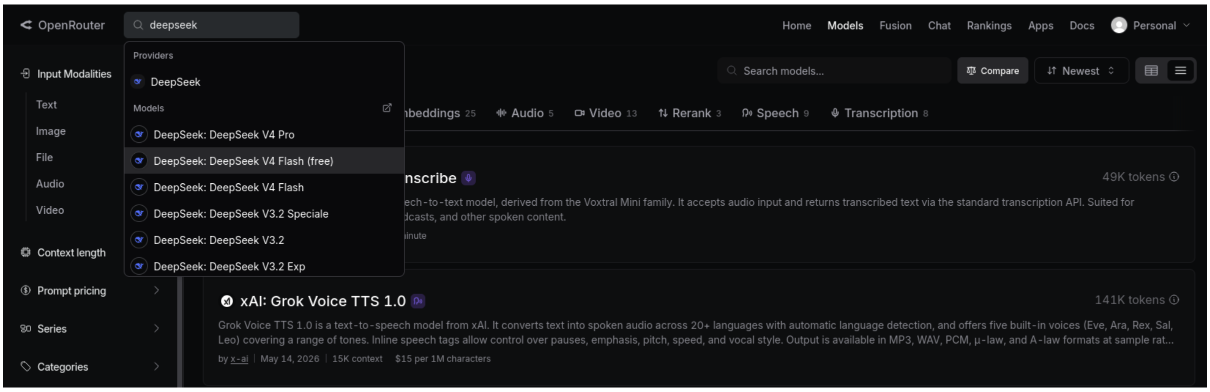
Step 5: Exploring a Free Model on OpenRouter

Before moving to the next practical, we will quickly test whether a free OpenRouter model responds properly to a cybersecurity-related prompt.

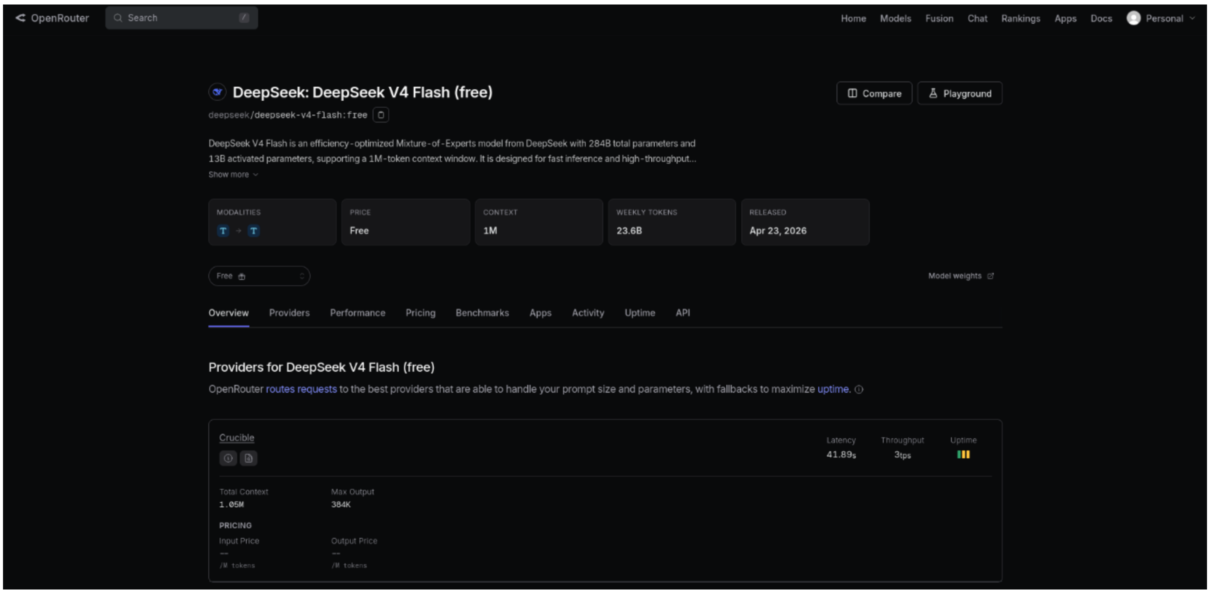
This step is only for basic testing. We will not upload files, logs, or network captures in this practical.

Action to Perform

1. On OpenRouter, go to the **Chat** option from the top navigation bar.
2. Search for deepseek.
3. Select whose name or model ID ends with (free)



4. You will get a screen like this:



What to Observe on the Model Details Page

After selecting the model, OpenRouter opens a detailed model information page.

1. Model Name and Model ID

At the top of the page, you can see the model name and its exact model ID.

2. Model Description

The short description explains what type of model it is and what it is designed for.

In this case, the page mentions that DeepSeek V4 Flash is optimized for efficiency and high throughput. This means it is intended to respond quickly and handle requests efficiently.

3. Modalities

The **Modalities** section shows what type of input and output the model supports.

This means the model accepts text prompts and returns text responses.

4. Price

The **Price** section shows:

Free

This confirms that the selected model is available as a free model on OpenRouter.

5. Context

The **Context** section shows how much text the model can handle in one conversation or request.

Here, the context value is:

1M

This indicates a large context window. A larger context window is useful when working with long prompts, long instructions, reports, logs, or large pasted text.

6. Weekly Tokens

The **Weekly Tokens** section shows recent usage volume for the model on OpenRouter.

This is not your personal usage. It indicates how much the model is being used across OpenRouter.

7. Released

The **Released** section shows the model release date.

This helps identify whether the model is recent or older.

Performance Section

The **Performance** section shows runtime behavior such as:

Metric	Meaning
Throughput	How fast the model generates output
Latency	Time taken before the model starts responding
E2E Latency	End-to-end response time
Tool Call Error Rate	Errors when the model is used with tools
Structured Output Error Rate	Errors when returning structured formats such as JSON

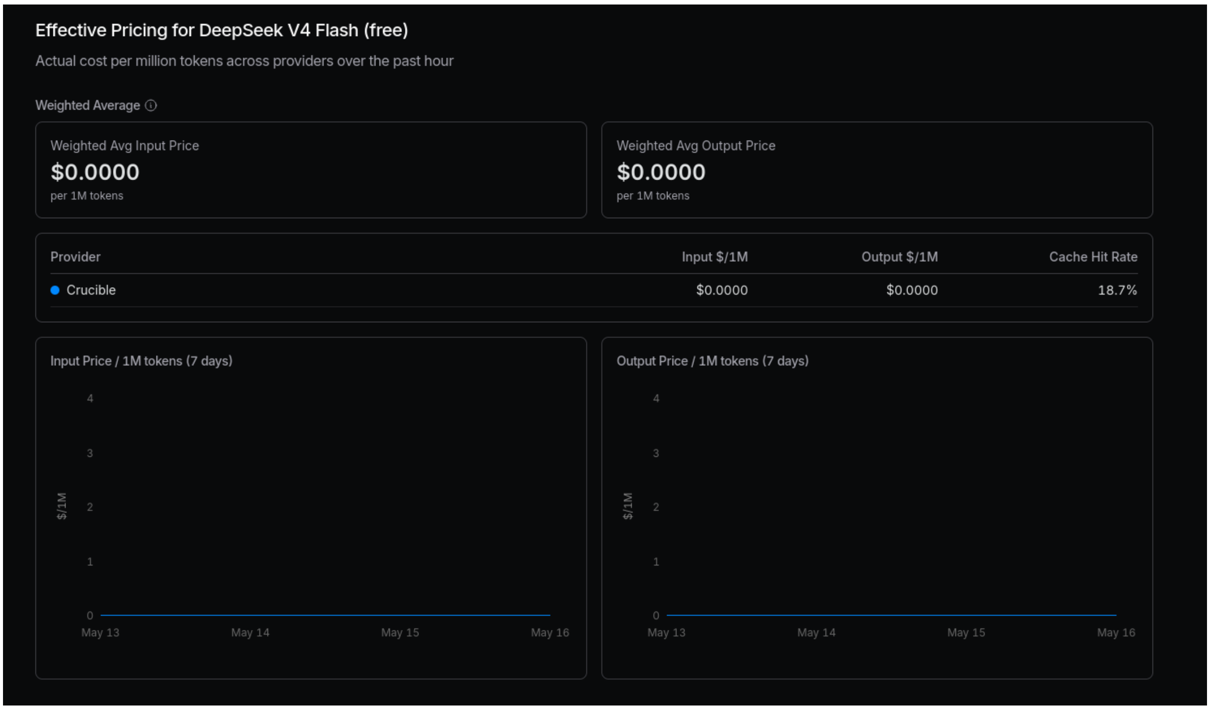
Pricing Section

The **Pricing** section confirms the actual input and output cost.

Here, both input and output pricing are:

$0.0000

This confirms that the selected model is free at the time of checking.



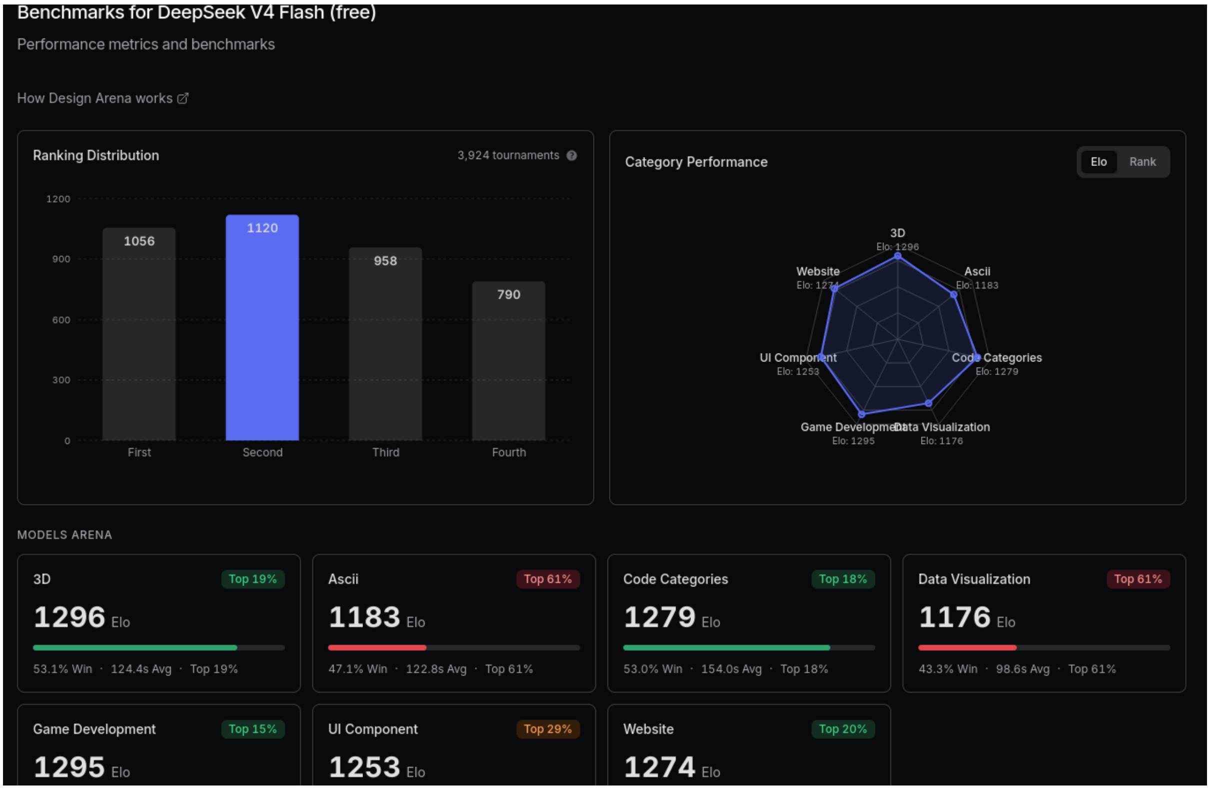
Benchmarks Section

The **Benchmarks** section shows how the model performs in different evaluation categories.

Examples shown include:

- 3D
- ASCII
- Code Categories
- Data Visualization
- Game Development
- UI Component
- Website

For this cybersecurity course, benchmarks are useful only as a general quality signal. A high benchmark score does not automatically mean the model is best for cybersecurity analysis, so we will test it using cybersecurity prompts.



Other Sections on the Same Page

You may also see the following sections further down the page:

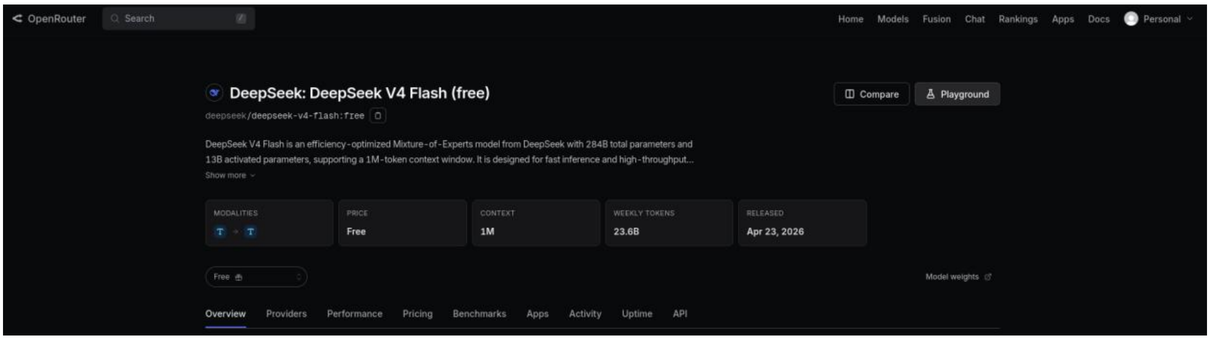
Section	Very Short Explanation
Apps using DeepSeek V4 Flash (free)	Shows applications or integrations currently using this model
Recent activity on DeepSeek V4 Flash (free)	Shows total usage per day on OpenRouter
Uptime stats for DeepSeek V4 Flash (free)	Shows reliability/availability statistics for the provider serving this model

Step 6: Navigate from Here to Test a Prompt

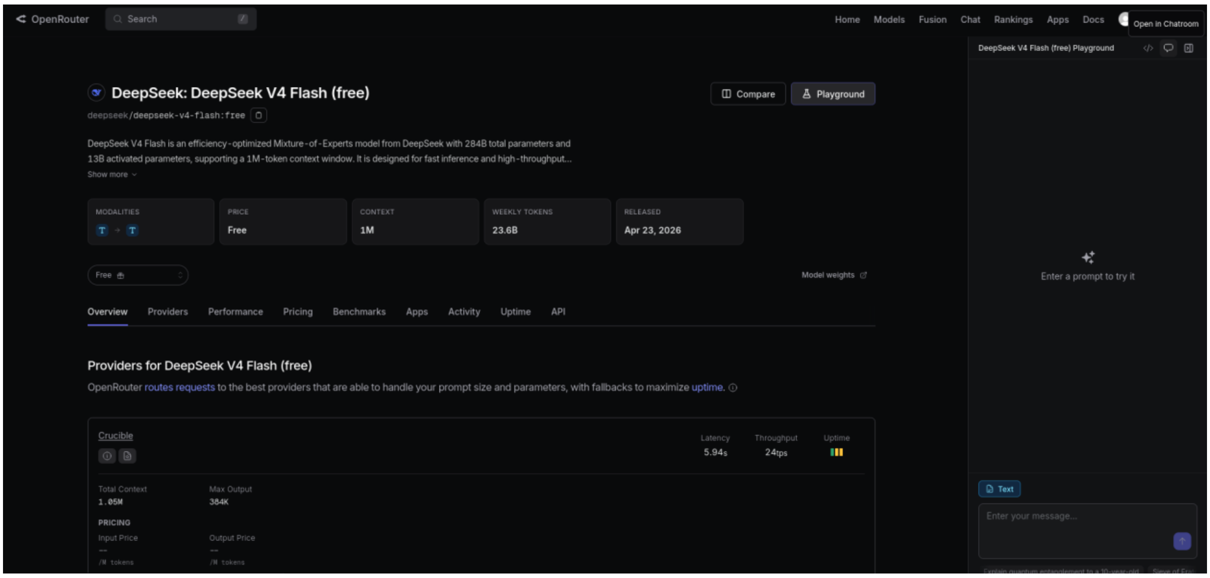
To test the model, use the **Playground** button.

Action to Perform

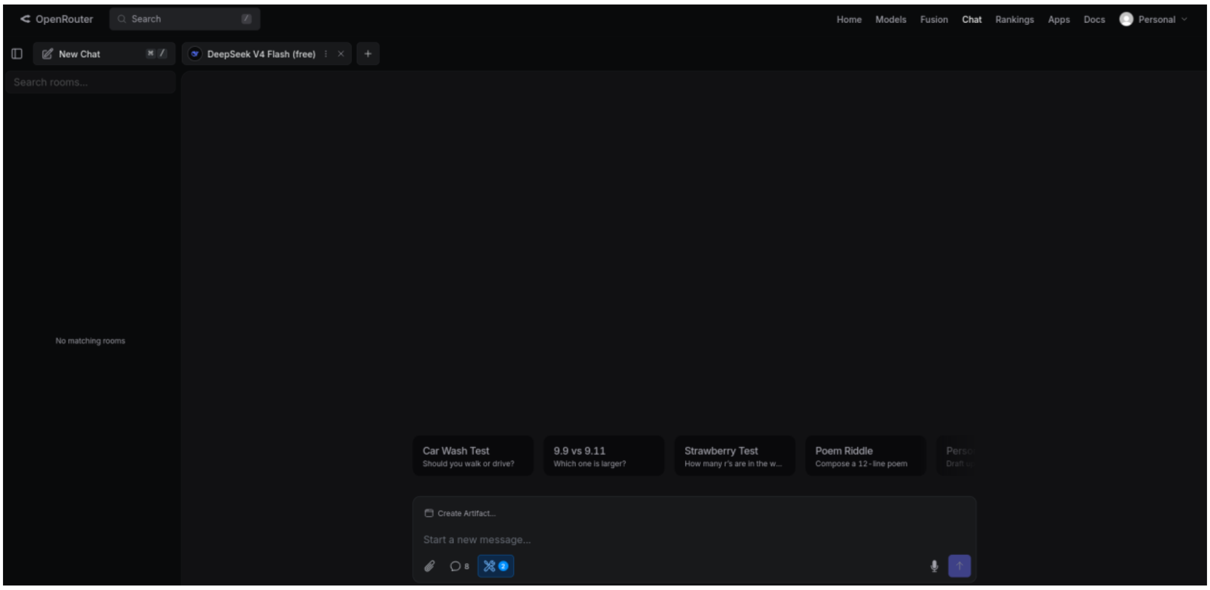
1. Stay on the DeepSeek V4 Flash model page.
2. Click the **Playground** button near the top-right of the page.



3. OpenRouter will open a chat/playground interface with this model selected.



4. You can select “Open in Chatroom” option here.



5. In the prompt box, enter the cybersecurity test prompt.

   “You are a cybersecurity analyst.

   Explain what a phishing email is in simple terms.

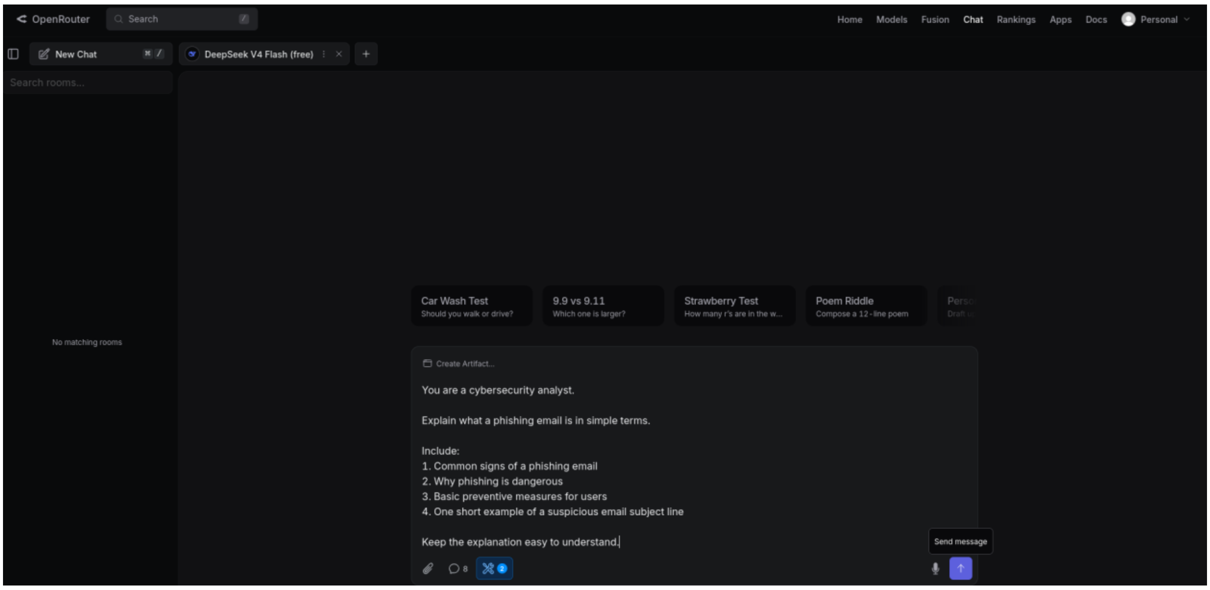
   Include:

   1. Common signs of a phishing email

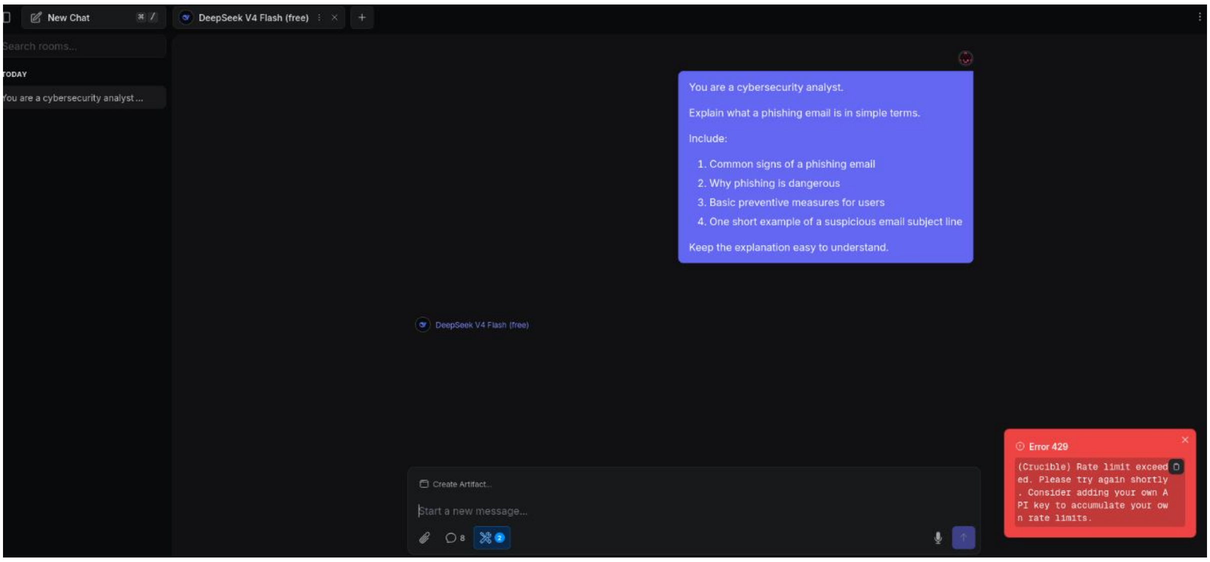
   2. Why phishing is dangerous

3. Basic preventive measures for users
4. One short example of a suspicious email subject line

Keep the explanation easy to understand.”



6. Submit the prompt by clicking on send message up arrow icon.
7. Observe the model response.



You may encounter the error message:

Error 429

(Crucible) Rate limit exceeded.

Please try again shortly.

Consider adding your own API key to accumulate your own rate limits.

This means the **provider serving this model**, in this case **Crucible**, has hit a rate limit for the free/shared route.

Why This Happens on the First Message

OpenRouter free models often run through shared provider capacity. So even if this is your first message, many other users may already be using the same free model/provider at the same time.

Step 7: Troubleshooting Rate Limit Error

Option 1: Wait and Retry

Wait for a short time and try again.

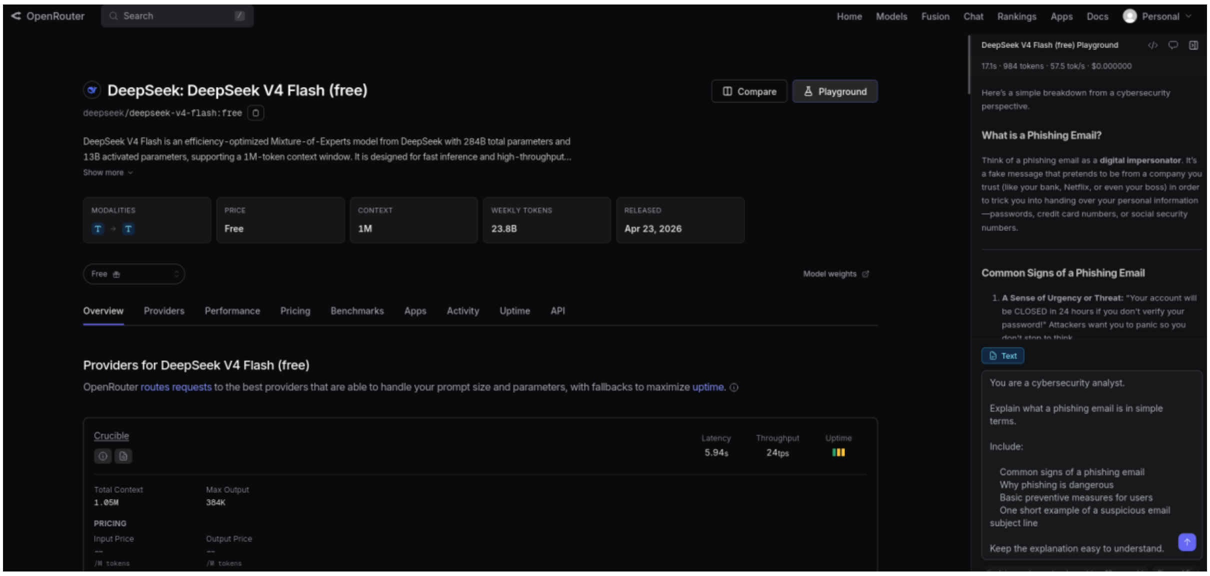
This is the simplest option.

Option 2: Use Chat Playground Sidebar Instead of opening in chatroom

From the top navigation bar, click:

Chat

Select the deepseek(free) model again



Option 3: Try Another Free Model

Go back to **Models**, search for another model ending in (free)

Option 4: Add an OpenRouter API Key Later

The message says:

Consider adding your own API key to accumulate your own rate limits.

This means API-key-based usage may have separate limits compared to shared playground usage.

For this first practical, learners can simply switch to another free model.